Take Home Assignment

Telemetry & Product Data Analytics Assessment

Overview

You are provided with multiple datasets representing operational telemetry, session activity, usage behavior, battery behavior, and machine fault events collected from connected consumer devices.

The datasets are from a single product family and represent anonymized machine populations across multiple regions.

Your objective is to:

- Understand the available datasets
- Assess data quality
- Generate meaningful insights
- Communicate findings through visualizations
- Recommendations for improving data quality and analyses

Datasets Provided

Dataset 1 – Cleaning Sessions

Contains cleaning activity information recorded during machine operation.

Example Fields

Field	Description
anonymized_serial	Anonymous machine identifier
date	Session timestamp
region	Geographic region
session_id	Session identifier
clean_duration	Duration of cleaning activity

battery_charge_start	Battery level at start
battery_charge_end	Battery level at end
battery_health	Battery health score
battery_temperature_hot	Hot temperature flag
battery_temperature_cold	Cold temperature flag
battery_temperature_ok	Normal temperature flag
battery_soc_state	Battery charge state
model	Product model
dust particle metrics	Collected particulate measurements

Source:

Clean Sessions_Normal Machines.csv

Clean Sessions_Return/Repair Machines.csv

Dataset 2 – Motor Telemetry

Contains operating-condition distributions recorded by the machine.

Available Metrics

- Speed distributions
- Power distributions
- Pressure distributions
- Exhaust temperature distributions
- Ambient temperature
- Ambient pressure
- Cumulative runtime

Source:

Motor_Normal Machines.csv

Motor_Return/Repair Machines.csv

Dataset 3 – Battery Telemetry

Contains battery behavior and health metrics.

Available Metrics

- Cell voltages
- State of Charge (SoC)
- State of Health (SoH)
- Charge behavior
- Discharge behavior
- Runtime
- Battery temperature
- Battery resistance estimates
- Charge and discharge statistics

Source:

Battery_Normal Machines.csv

Battery_Return/Repair Machines.csv

Dataset 4 – Fault Events

Contains machine fault and alarm events.

Example Fields

Field	Description
anonymized_serial	Anonymous machine identifier
session_id	Session identifier
fault_type	Fault category
fault_value	Fault state/value
fault_timestamp	Event timestamp
product_alarm_source	Source subsystem
hex_error_code	Fault code
product_alarm_fault_severity	Severity level
fault_status	Raised / Cleared

Fault Types & Values

Attribute ID	Fault Type	Fault Value	Meaning
12	Airway Blockage	0	Unit is not blocked
12	Airway Blockage	1	Unit inlet is blocked but motor is still running
12	Airway Blockage	2	Unit inlet is blocked and in error
12	Airway Blockage	3	Unit outlet is blocked and in error
7	Brush Bar Stall	N/A	Brush bar stall detected
11	Battery Temperature	0	Battery temperature OK
11	Battery Temperature	1	Battery temperature COLD
11	Battery Temperature	2	Battery temperature HOT
13	Filter Wash	0	Filter is OK
13	Filter Wash	1	Filter needs to be cleaned

Source:

Faults_Normal Machines.csv

Faults_Return/Repair Machines.csv

Important notes regarding telemetry data

1) Several fields within the motor telemetry and battery telemetry datasets represent cumulative machine lifetime counters rather than period-level measurements. And they are captured on a weekly basis.

For example, a value reported on Wk 30 may already include all usage accumulated during Wks 1-29.

Candidates should assess whether cumulative counters require transformation (for example, delta calculations between observations) before performing trend analysis or feature engineering.

Any assumptions made regarding cumulative metrics should be documented.

2) Candidates may combine the Normal Machines and Return/Repair Machines datasets into a single analysis population. Distinguishing between the two groups is optional.

Supporting Reference Files

Motor Telemetry Bin Definitions

Provides definitions for:

- Speed ranges
- Power ranges
- Pressure ranges
- Temperature ranges

These files describe how telemetry distributions are encoded.

Source:

DDM_Bins.json

Battery Telemetry Bin Definitions

Provides definitions for:

- Charge duration ranges
- Discharge duration ranges
- Voltage bands
- Temperature bands
- Charge and discharge operating regions

Source:

BMS_Bins.json

Deliverables

Part 1 – Data Understanding & Quality Assessment

Identify and discuss the most significant data quality issues observed. Some examples below:

- Missing data
- Duplicate records
- Invalid values
- Outliers
- Timestamp consistency and other issues
- Potential risks or limitations

Part 2 – Unified Analytical Model

Describe how you would join and organize the datasets for analysis.

You may include:

- Data model
- Entity relationship diagram
- Star schema
- Lakehouse/Warehouse design

Briefly describe how you would productionize this analysis, including high-level ingestion, transformation, storage, and visualization layers.

Part 3 – Exploratory Analysis

Identify and explain 3-4 meaningful insights that you believe are most valuable to product, engineering, or reliability stakeholders.

Insights may be drawn from one or more of the following areas:

Usage Behaviour

- Session duration trends
- Regional differences
- Model differences
- Heavy vs light usage patterns

Product Performance

- Runtime behavior
- Power consumption patterns
- Temperature behaviour
- Usage distributions

Battery Behaviour

- Battery health distribution
- Charge/discharge behaviour
- Potential degradation indicators

Fault Analysis

- Common fault codes
- Fault frequency
- Severity distribution
- Repeat fault patterns
- Fault co-occurrence

Part 4 – Visualization & Storytelling

Create a dashboard or report showing key findings with screenshots of the plots

Expected audience:

- Product managers
- Quality & Reliability engineers
- Engineering leadership

Preferred tools:

- Tableau
- Power BI

- Streamlit
- Any equivalent analytics tool

Expected Outcome

There is no single "correct" answer.

We are interested in:

- How you structure and combine ambiguous datasets
- How you identify limitations
- How you design and organize data models to support analysis
- How you derive useful insights
- How you communicate findings
- How you think about future analytical opportunities from the available data

Submission Requirements

1. Analysis Report

Submit a report (5 slides or 5 pages) covering:

Executive Summary

- Key findings
- Most important insights
- Business and engineering implications

Data Understanding

- Description of datasets
- Assumptions made
- Relationships identified between datasets

Data Modelling & Data Preparation

- Data relationships and join strategy
- Assumptions required to combine datasets
- Recommended analytical data model
- Key transformation steps

- How the data could be structured for reporting and future analytics

Data Quality Assessment

- Missing data
- Duplicate records
- Invalid values
- Outliers
- Data limitations
- Recommendations for improving data quality

Analytical Findings

- 3-4 insights from these categories
    - Usage behaviour insights
    - Fault analysis insights
    - Battery performance insights
    - Product performance insights
- Any additional observations

Recommendations

- Suggested next analyses
- Data improvements that would enable better analytics

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2. Code Repository (optional)

Submit a GitHub repository (or zipped source folder) containing:

- Data processing code
- Data quality checks
- Analysis notebooks/scripts
- Feature engineering logic
- Dashboard code (if applicable)

A clear README should describe:

- Approach
- Assumptions
- How to run the analysis

3. Dashboard / Visualization

Create a lightweight dashboard using a tool of your choice and provide an accessible link if available (preferred)

Alternatively, provide screenshots of the relevant insights with explanations of the plots

Examples:

- Tableau
- Power BI
- Streamlit
- Python-based

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4. Assumptions & Limitations

Document:

- Data limitations encountered
- Assumptions made during analysis
- Additional data that would improve confidence in conclusions
- Known gaps that prevent deeper insights

What We Value

We are looking for candidates who can:

- Structure ambiguous datasets
- Identify data quality issues
- Recommend logical data models
- Define practical data preparation and transformation approaches
- Generate meaningful insights
- Communicate findings clearly
- Explain trade-offs and assumptions